THE UNIVERSITY
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Homogeneity Fusion on Sparse Grouped Network VAR Models

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The Model

- Directed network with adjacency matrix $A = (a_{ij})_{1\le i,j\le d}$.
- Latent partition $\mathcal{G} = \{G_1, \dots, G_K\}$ of $[d]$ into K groups. We denote the group membership of node i by g_i.

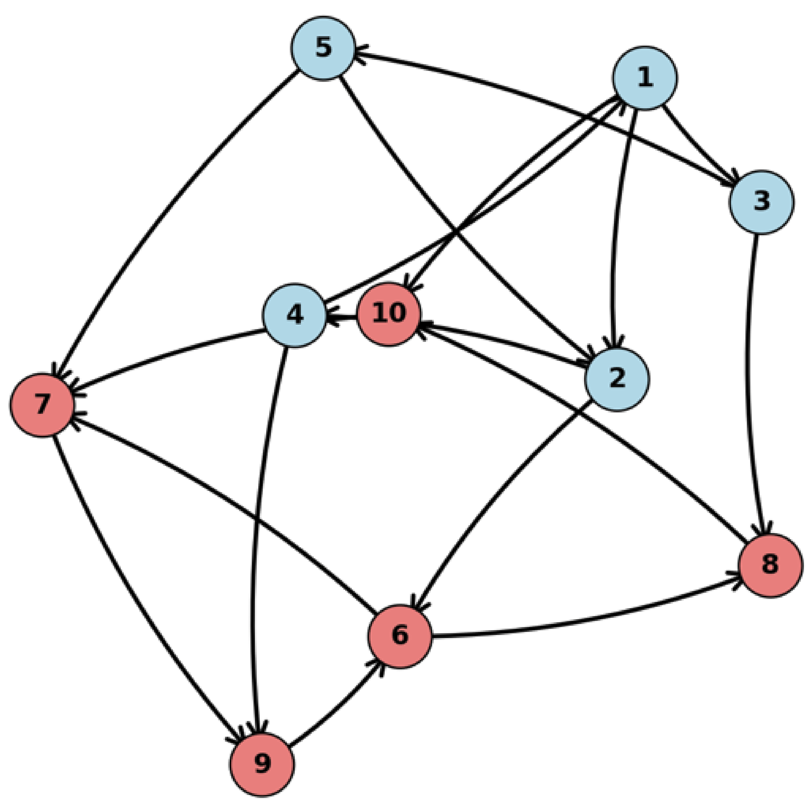

Figure 1: Directed network with node set clustered in 2 groups.

- GNAR model:

$$Y_{i,t} = \sum_{j=1}^{d} \phi_{g_i g_j} \frac{a_{ij}}{n_i} Y_{j,t-1} + \nu_{g_i} Y_{i,t-1} + \varepsilon_{i,t}, \quad i \in [d],\ t \in [T],$$

where $n_i = \sum_{j=1}^{d} a_{ij}$ the out-degree of i, and $\varepsilon_{i,t} \overset{\text{i.i.d.}}{\sim} N(0,\sigma^2)$.

- Stationarity condition: $\max_{k\in[K]} \left(\max_{l\in[K]} |\phi_{kl}| + |\nu_k|\right) < 1$

Homogeneity fusion estimator

- *Goal:* estimate unknown network A, groups $\mathcal{G}$, and parameters Φ, ν.
- *Optimization problem:* minimize ℓ_2-error subject to sparsity constraint:

$$\underset{\substack{A\in\{0,1\}^{d\times d}, \mathcal{G}\in[K]^d \\ \Phi\in\mathbb{R}^{K\times K}, \nu\in\mathbb{R}^K}}{\arg\min} \sum_{t=1}^{T}\sum_{i=1}^{d} \left(Y_{i,t} - \sum_{j=1}^{d} \phi_{g_i g_j} \frac{a_{ij}}{n_i} Y_{j,t-1} - \nu_{g_i} Y_{i,t-1} \right)^2$$

$$\text{s.t.} \quad \sum_{i=1}^{d}\sum_{j=1}^{d} a_{ij} \le s.$$

Main Theorems

1. *Misspecification probability:*

$$\mathbb{P}\left((\widehat{A}, \widehat{\mathcal{G}}) \nsim (A^*, \mathcal{G}^*) \right) \to 0, \quad \text{as } d, T \to \infty$$

2. *Estimation error:*

$$\|\widehat{\beta}_{\text{ol}} - \beta^*\|_2 \to 0 \quad \text{w.h.p,} \quad \text{as } d, T \to \infty,$$

where β is a vector including the diagonal coefficients ν_{g_i} and off-diagonal $\phi_{g_i g_j} \frac{a_{ij}}{n_i}$.

Proof ideas:

- Bound

$$\mathbb{P}\left((\widehat{A}, \widehat{\mathcal{G}}) \nsim (A^*, \mathcal{G}^*) \right) \le$$

$$\sum_{\substack{(\mathcal{G},A)\neq(\mathcal{G}^*,A^*) \\ |\mathcal{G}|=|\mathcal{G}^*|, \|A\|_0 \le \|A^*\|_0}} \mathbb{P}\left(\min_{\widetilde{\beta}:\, (\widetilde{\mathcal{G}}, \widetilde{A}) = (\mathcal{G}, A)} \|Y - Z\widetilde{\beta}\|_2^2 < \|Y - Z\widehat{\beta}_{\text{ol}}\|_2^2 \right).$$

- Rewrite as

$$\begin{aligned} &\mathbb{P}\left(\min_{\widetilde{\beta}:\, (\widetilde{\mathcal{G}}, \widetilde{A}) = (\mathcal{G}, A)} \|Y - Z\widetilde{\beta}\|_2^2 < \|Y - Z\widehat{\beta}_{\text{ol}}\|_2^2 \right) \\ &= \mathbb{P}\left(\|Y - Z\widehat{\beta}\|_2^2 < \|Y - Z\widehat{\beta}_{\text{ol}}\|_2^2 \right) \\ &= \mathbb{P}\left(\varepsilon'(P_{\mathbb{G}} - P_{\mathbb{G}^*})\varepsilon - 2\varepsilon'(I - P_{\mathbb{G}})Z\beta^* > \|(I - P_{\mathbb{G}})Z\beta^*\|_2^2 \right) \end{aligned}$$

- Bound using concentration inequalities like Hanson-Wright.

Implementation

1. Estimate the underlying network with one of two methods:
   - *Sure Independence Screening*: obtain a vector of marginal regression estimates, rank its elements in decreasing order, and retain a fixed proportion of them.
   - *Best Subset Selection*: select the subset of predictors that minimizes the residual sum of squares for a given subset size.
2. For a given network, simultaneously estimate the groups and parameters by solving the following **mixed integer program**:

$$\underset{\substack{B\in\mathbb{R}^{d\times d}, \Phi\in\mathbb{R}^{K\times K}, \\ \nu\in\mathbb{R}^K, \Omega\in\{0,1\}^{d\times K}}}{\arg\min} \sum_{t=1}^{T}\sum_{i=1}^{d} \left(Y_{i,t} - \sum_{j=1}^{d} b_{ij} Y_{j,t-1} \right)^2$$

$$\begin{aligned} \text{s.t.} \quad & \omega_{ik} \in \{0,1\}, \quad i \in [d], k \in [K] \\ & \sum_{k=1}^{K} \omega_{ik} = 1, \quad i \in [d] \\ & b_{ii} - \nu_k \le M(1 - \omega_{ik}), \quad i \in [d], k \in [K] \\ & b_{ii} - \nu_k \ge -M(1 - \omega_{ik}), \quad i \in [d], k \in [K] \\ & n_i b_{ij} - \phi_{kl} a_{ij} \le M(2 - \omega_{ik} - \omega_{jl}), \quad i, j \in [d], i \neq j,\ k, l \in [K] \\ & n_i b_{ij} - \phi_{kl} a_{ij} \ge -M(2 - \omega_{ik} - \omega_{jl}), \quad i, j \in [d], i \neq j,\ k, l \in [K] \end{aligned}$$

3. Iterate the previous steps.

Applications

- *FRED-MD dataset:* 134 monthly U.S. macroeconomic variables.
- We focus on a subset of 42 variables in two groups:
  - Group 6: Interest & Exchange Rates (22 nodes)
  - Group 7: Prices (20 nodes)
- We treat variables as nodes in a network and apply our method to estimate:
  - The edge structure (adjacency matrix)
  - Group memberships

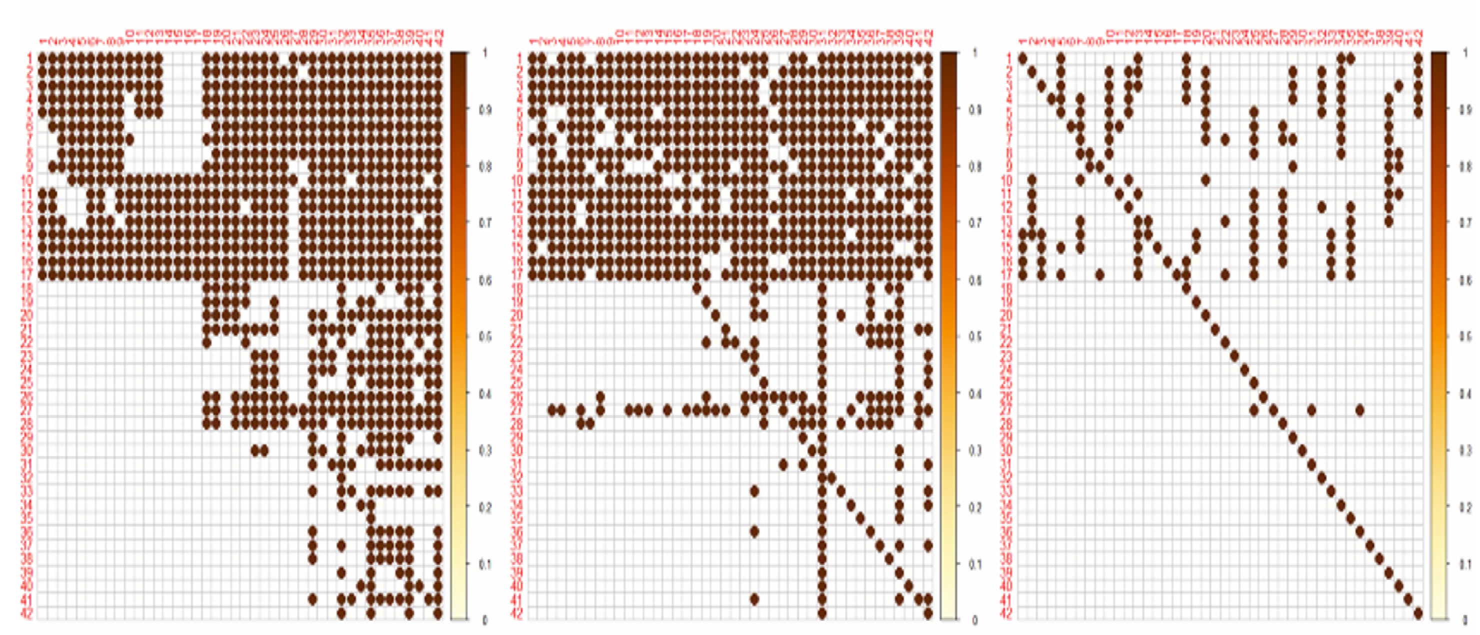

Figure 2: Estimated adjacency matrices. Left: Homogeneity Fusion, Middle: Best Subset, Right: Adaptive Lasso